

Proposed Use Cases and Feasible Implementation

The LiDAR sensor is mounted approximately 14 cm above the ground and operates as an observation-only component. It does not influence motion control, safety decisions, or real-time behavior of the robot. Since ROS Melodic and a catkin workspace are already available on the platform, LiDAR data can be acquired, logged, and analyzed using standard ROS tooling without modifying the existing control scripts.

Environment Mapping and Spatial Context Logging

LiDAR data can be used to capture two-dimensional representations of the robot's surroundings during experimental runs. At a mounting height of 14 cm, the sensor primarily observes vertical structures such as walls, furniture, table legs, cabinets, and other obstacles that are relevant for indoor navigation. Floor texture and ground markings are intentionally excluded, which aligns well with the camera-based line-following task. Implementation is straightforward using the existing `rplidar_ros` node, which publishes full 360° LaserScan data. These scans can be recorded using `roslaunch` during experiments and processed offline to visualize or reconstruct the spatial layout of test environments. This allows experimental conditions to be documented and compared across different rooms or laboratories without affecting robot behavior.

Ground-Truth Reference for Obstacle Detection

The LiDAR can serve as a reference sensor for evaluating the ultrasonic obstacle detector. While the ultrasonic sensor provides a single frontal distance measurement close to the ground, the LiDAR at 14 cm height captures a wider spatial profile of obstacles at a higher plane. This difference in sensing height is useful for identifying cases where the ultrasonic sensor triggers due to small ground-level objects or fails to detect obstacles that are only partially visible at lower heights.

In practice, LiDAR scans and ultrasonic readings can be time-stamped and logged independently. Offline analysis can then compare LiDAR distance measurements in the forward direction with ultrasonic trigger events to identify false positives, false negatives, or delayed responses.

Fault Injection and Sensor Failure Analysis

LiDAR integration is particularly useful for fault injection experiments. During simulated or induced failures of the ultrasonic sensor, such as constant readings, noise injection, or delayed echoes, the LiDAR continues to provide an independent perception of the environment at a fixed height. This enables analysis of how the system behaves under degraded sensing conditions while retaining a reliable reference for obstacle presence and distance.

Because the LiDAR does not interact with the control logic, these experiments can be conducted without compromising system safety.

Post-Event and Stop-Condition Analysis

When the robot enters a stopped state due to the ultrasonic safety threshold, LiDAR data can be used to analyze the cause of the stop. At a 14 cm mounting height, the LiDAR can help determine whether the obstacle was a wall, a human leg, furniture, or another vertical object, and whether it was static or moving.

This can be implemented by extracting LiDAR scans around the time of the stop event and examining obstacle distance, angular extent, and relative position. Such analysis helps classify stop events and improves understanding of system behavior without altering runtime logic.

Correlation with Vision-Based Inference

LiDAR data can be correlated with outputs from the TensorRT-based object detection pipeline to enable multimodal analysis. Although inference does not control movement, combining visual detections with LiDAR geometry allows validation of detected objects against physical distance and spatial structure.

The LiDAR's mounting height is especially relevant here, as it often intersects with the lower portion of detected objects such as people or furniture, providing complementary spatial information to the camera. Synchronization can be performed offline using timestamps from recorded camera frames, inference outputs, and LiDAR scans.

Analysis of Side and Rear Environment During Motion

Because the robot performs turns and rotations during line following, full 360° LiDAR scans provide information about the side and rear environment that is not captured by the camera or ultrasonic sensor. At 14 cm height, this data reflects constraints imposed by surrounding structures during motion, such as narrow corridors or nearby walls.

This information can be used to analyze how environmental geometry influences robot behavior indirectly, particularly during turns or line-loss events, without requiring any real-time integration.

Validation of Safety Margins

The LiDAR can be used to evaluate the effectiveness of the current ultrasonic stop threshold of 4 cm. By examining LiDAR-measured distances at the moment ultrasonic stops occur, it becomes possible to assess whether the safety margin is conservative, insufficient, or overly restrictive in different environments.

Although LiDAR measurements are taken at a higher plane than the ultrasonic sensor, they still provide useful context for understanding obstacle proximity and validating safety assumptions.

Long-Term Data Collection and Platform Extension

LiDAR data can be collected continuously over long periods for offline experimentation. Recorded datasets can later be reused for mapping, SLAM, or localization experiments without modifying the current control system. This approach allows future research and development while preserving the deterministic behavior of the existing platform.